

Glory Kellogg

Applied Mathematics Honors Student

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Education

Illinois Institute of Technology, Chicago, IL

January 2026 – Present

B.S. Applied Mathematics, Expected December 2028 | GPA: 4.0/4.0

Honors Program | Minor in Computer Science

Relevant Coursework: Linear Algebra, Discrete Mathematics, Real Analysis, Probability

Triton College, River Grove, IL | GPA: 4.0/4.0

August 2024 – December 2025

Phi Theta Kappa Honor Society

Relevant Coursework: Calculus II, Calculus III, Differential Equations, Physics I

Research Interests

Mathematical biology | Pattern formation | Differential equations | Graph theory | Computational modeling

Technical Skills

- Programming: Python | C++ | Java | C
- Tools: Git | GitHub | LaTeX
- Web & Publishing: HTML | CSS | JavaScript | Quarto | GitHub Pages
- Design: Adobe Illustrator | Adobe Photoshop

Independent Research

Graph-Based Models of Morphogen Diffusion in Butterfly Wing Development

- Investigating mathematical models of butterfly wing pattern formation through geometry, graph representations, and computational modeling.
- Developing computational methods to extract wing structure from photographs for comparison with mathematical models.
- Raising and documenting painted lady butterflies while studying developmental biology and pattern formation.

Nonlinear Preferential Attachment in Collaboration Networks

- Developing an agent-based model of collaboration networks to study the emergence of structural inequality through nonlinear preferential attachment.
- Designed Monte Carlo experiments to investigate the effects of attachment bias on network growth and degree inequality.

Leadership Experience

Social Media Manager | *SIAM IIT Student Chapter* (May 2026 – Present)

- Lead digital communications and outreach for Illinois Tech's Society for Industrial and Applied Mathematics student chapter.
- Design promotional materials for seminars, workshops, and chapter events using Adobe Illustrator and Photoshop.
- Designed and developed the chapter website using HTML, CSS, JavaScript, Quarto, Git, and GitHub Pages.
- Represented Illinois Tech at the 2026 SIAM Annual Meeting (AN26) in Cleveland.

Work Experience

Founder & Owner | *Sapphaux Ceramics* (June 2019 – July 2023)

- Built and maintained an e-commerce website using Squarespace and custom CSS.
- Created systems for inventory management, order fulfillment, and sales tracking.
- Managed day-to-day operations while coordinating with customers and retail partners.

Manager on Duty | *The Shop at Equinox* (November 2022 – March 2023)

- Promoted from Sales Associate to Manager on Duty within one month.
- Directed daily store operations, inventory management, scheduling, and sales reporting.
- Trained and supervised retail team members.

Bank Teller | *Old Second National Bank* (December 2020 – April 2021)

- Processed financial transactions with accuracy and compliance.
- Balanced cash drawers and resolved customer inquiries.

Host | *Scofflaw* · *Handlebar* · *Sink/Swim*

- Managed reservations and customer flow in high-volume restaurants.
- Coordinated communication between front- and back-of-house staff.

Sales Associate | *Madewell* · *Fringe, A Salon Inc.* · *Steel Petal Press*

- Managed inventory, fulfilled customer orders, and maintained sales and client records.
- Delivered customer service in fast-paced retail and hospitality settings.